

Experiment 6 — Fine-Tuning and Comparative Evaluation of a Pretrained BERT Model for Sentence Classification

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Generative AI (PE III) — CE413 | Semester VII
SMS Spam/Ham Classification | bert-base-uncased
NVIDIA GeForce RTX 4050 Laptop GPU | CUDA 13.0

1. Aim

To design, implement, fine-tune, and evaluate a pretrained BERT model for a real-world sentence classification task and analyze preprocessing, tokenization, sequence length, classification-head architecture, hyperparameters, class imbalance, and input robustness.

2. Problem Statement

A BERT-based SMS classifier was implemented using Hugging Face Transformers. Messages are classified as Ham (0) or Spam (1), followed by controlled comparisons and robustness analysis.

3. Dataset and Preprocessing

Measure	Before cleaning	After cleaning
Records	5,574	5,171
Duplicate rows	403	0
Missing text	0	0
Ham (0)	4,827	4,518
Spam (1)	747	653
Average characters	81.48	80.33

Final split: Training = 3,619; Validation = 776; Testing = 776. Two preprocessing variants were evaluated: minimal preprocessing and cleaned/normalized text.

4. Tokenization

The tokenizer was bert-base-uncased using WordPiece tokenization. The experiment inspected tokens, token IDs, attention masks, padding and truncation.

Maximum length	Observed non-padding token counts
32	26, 32, 32, 17, 32
128	26, 33, 32, 17, 35

5. Baseline BERT

Metric	Value
Accuracy	98.45%
Precision	95.74%
Recall	91.84%

F1	93.75%
Macro-F1	96.43%
Validation loss	0.0548
Training time	75.43 s

6. Comparative Results

Experiment	Seq.	LR	Head	Weighted	Accuracy	Precision	Recall	F1	Macro-F1	Time (s)
E1_baseline_bert	32	2e-05	baseline	No	0.9845	0.9574	0.9184	0.9375	0.9643	75.43
E2_preprocessing_normalized	32	2e-05	baseline	No	0.9832	0.9293	0.9388	0.9340	0.9622	75.83
E3_sequence_128	128	2e-05	baseline	No	0.9897	0.9592	0.9592	0.9592	0.9766	191.87
E4_hyparam_1	32	2e-05	baseline	No	0.9832	0.9293	0.9388	0.9340	0.9622	77.35
E4_hyparam_2	32	3e-05	baseline	No	0.9794	0.9020	0.9388	0.9200	0.9541	76.18
E4_hyparam_3	32	5e-05	baseline	No	0.9845	0.9300	0.9490	0.9394	0.9653	76.53
E5_head_linear	32	2e-05	linear	No	0.9858	0.9485	0.9388	0.9436	0.9677	76.48
E5_head_dropout	32	2e-05	dropout	No	0.9832	0.9208	0.9490	0.9347	0.9625	76.58
E5_head_dense_dropout	32	2e-05	dense_dropout	No	0.9807	0.9029	0.9490	0.9254	0.9571	77.13
E6_imbalance_standard	32	2e-05	linear	No	0.9781	0.8932	0.9388	0.9154	0.9514	75.67
E6_imbalance_weighted	32	2e-05	linear	Yes	0.9820	0.9118	0.9490	0.9300	0.9598	75.59

7. Sequence Length Analysis

Length 32 baseline: accuracy 98.45%, F1 93.75%, macro-F1 96.43%, training time 75.43 s. Length 128: accuracy 98.97%, F1 95.92%, macro-F1 97.66%, training time 191.87 s. Peak GPU memory was not recorded.

8. Hyperparameter Analysis

Learning rates 2e-5, 3e-5 and 5e-5 were compared with the other principal settings held fixed. Their test F1 values were 0.9340, 0.9200 and 0.9394. The reported sweep did not independently vary batch size, epochs and weight decay.

9. Classification Head Comparison

Linear, Dropout and Dense+Dropout heads achieved test F1 values of 0.9436, 0.9347 and 0.9254 respectively. The modified heads did not provide a consistent test-set improvement.

10. Class Imbalance Analysis

Approach	Accuracy	Precision	Recall	F1	Macro-F1	Confusion matrix
Standard CE	0.9781	0.8932	0.9388	0.9154	0.9514	[[667,11], [6,92]]
Weighted CE	0.9820	0.9118	0.9490	0.9300	0.9598	[[669,9],[5,93]]

Class weighting improved spam recall, spam F1 and macro-F1 in this run.

11. Robustness and Error Analysis

Twelve manually constructed examples were tested. Eleven were correct and one was incorrect. The only error was 'congrts u won a cash prize call now!!!', expected Spam but predicted Ham. The supplied paraphrased spam example was correct.

The manual asks for at least 10 incorrectly classified examples. This run produced only one, so no additional errors are fabricated. A larger adversarial test set would be needed to satisfy that requirement literally.

12. Training Behaviour

Epoch	Train loss	Validation loss	Validation accuracy	Validation Macro-F1
1	0.1770	0.0606	0.9858	0.9686
2	0.0361	0.0521	0.9871	0.9713
3	0.0070	0.0548	0.9858	0.9683

Training loss decreased strongly while validation loss remained near 0.05, showing strong training-set fitting without continuous validation improvement.

13. Answers to Results and Discussion Questions

i. BERT tokenization vs traditional preprocessing: Traditional NLP commonly applies manually designed operations such as stemming, lemmatization and stop-word removal. BERT uses WordPiece subword tokenization with a pretrained vocabulary, special tokens, token IDs, attention masks, padding and truncation.

ii. Role of [CLS]: [CLS] is placed at the beginning of the sequence and its final hidden representation is used as the sequence-level representation for classification.

iii. Sequence length: Length 128 improved accuracy from 98.45% to 98.97% and F1 from 93.75% to 95.92%, while training time increased from 75.43 s to 191.87 s.

iv. Hyperparameter effect: Learning-rate changes produced measurable but modest differences. The run did not independently sweep all suggested training parameters, so a definitive greatest-effect hyperparameter is not supported.

v. Classification-head generalization: The linear head produced F1 0.9436 and macro-F1 0.9677, higher than the tested Dropout and Dense+Dropout heads.

vi. Class imbalance: Weighted loss improved accuracy 97.81% → 98.20%, spam recall 93.88% → 94.90%, spam F1 91.54% → 93.00%, and macro-F1 95.14% → 95.98%.

vii. Frequent errors: Only one error occurred in the supplied robustness set, involving a heavily misspelled/informal spam message.

viii. Spelling/paraphrase robustness: The paraphrased example was correct; the heavily misspelled example was incorrect. This small test indicates sensitivity to spelling corruption.

ix. Best performance vs generalization: Validation loss and test performance were not always aligned; Dense+Dropout had lower validation loss than Linear but lower test F1.

x. Limitations: Strong class imbalance, a small robustness set, only one observed robustness error, incomplete hyperparameter sweep, no peak-memory measurement, changing SMS language, computational cost, and lack of external context for ambiguous messages.

14. Conclusion

The BERT sentence-classification pipeline was successfully implemented and evaluated. Results show strong baseline performance, improved test metrics with sequence length 128 at higher computational cost, no consistent benefit from complex classification heads, improved minority-class metrics with weighted loss, and sensitivity to severe spelling corruption. Conclusions are based on the supplied experimental outputs.

15. Required Plots

Generated files: plot1_training_loss.png, plot2_validation_loss.png, plot3_accuracy_f1.png, and plot4_confusion_matrix.png in results/plots/.

16. Repository

<https://github.com/CodeCraftsmanRaj/GenAI/tree/main/Exp6>

17. Execution

```
uv add -r requirements.txt
uv run main.py
```